

Pure Substances and Mixtures — Paper A

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, your answer must say whether the atoms are **chemically joined** or **not joined**, and whether there is **one kind of particle** or more than one.
- The mark for each question is shown in square brackets on the right.
- Reminder: a pure substance melts **sharply** at one temperature; an impure one melts **over a range**, and lower.

SECTION 1 – THE THREE WORDS

1. Complete each definition.

(a) An **element** contains only one kind of _____. [1]

(b) A **compound** contains two or more kinds of atom that are _____ together. [1]

(c) A **mixture** contains two or more _____ that are not chemically joined. [1]

2. Sort these six substances into the table: **oxygen, sea water, water, air, iron, carbon dioxide**. Write each name in one column only. [6]

Element	Compound	Mixture

3. State what is meant by a **pure substance** in chemistry, and explain why a carton labelled "pure orange juice" is not pure in this sense. [2]

- 4.** In the space below, draw a particle diagram for each of the following. Use circles of different shading for different kinds of atom, and label each box.

(a) An element made of single atoms. [1]

(b) A compound. [1]

(c) A mixture of two elements. [1]

(d) A mixture of an element and a compound. [1]

- 5.** A box contains particles. Every particle is a group of one large shaded circle joined to two small white circles, and every group is identical.

(a) How many kinds of atom are present? [1]

(b) Is the contents of the box an element, a compound or a mixture? Give a reason. [2]

6. Give **two** properties of a mixture that are different from the properties of a compound. [2]

7. Iron filings and sulfur powder are stirred together in a dish.

- (a) State what you would see, and how you could separate the two again. [2]

- (b) The dish is heated strongly and a single grey solid forms. State whether this solid is a mixture or a compound, and give one piece of evidence. [2]

8. Air is a mixture of gases. State the two main gases in air and explain why air is classed as a mixture rather than a compound. [3]

9. Give one example of a mixture of two **solids** that is useful, and say what it is used for. [3]

End of paper. Check every answer has a unit or a form the question actually asked for.